A Bounded Collapsed Almost Uncollapsed Map

Automated discovery packet

June 14, 2026

Source question

Braga's paper [1] asks Problem 2.6: "Is there a map $X \to Y$ which is collapsed, almost uncollapsed and bounded (in particular not solvent), for some Banach spaces X and Y?"

Hence, $\|b(2^{2^{k}})\| \to 0$, as $k \to \infty$. So, b is collapsed. $\square$

Problem 2.6. Is there a map $X \to Y$ which is collapsed, almost uncollapsed and bounded (in particular not solvent), for some Banach spaces X and Y?

Definitions used

For a map $f : (M, d) \to (N, \partial)$, the source defines

$$\rho_f(t) = \inf\{\partial(f(x), f(y)) : \ d(x, y) \geq t\}$$

and

$$\overline{\rho}_f(t) = \inf\{\partial(f(x), f(y)) : \ d(x, y) = t\}.$$

The map is uncollapsed if $\rho_f(t) > 0$ for some $t > 0$, and almost uncollapsed if $\overline{\rho}_f(t) > 0$ for some $t > 0$. In the terminology of the paper, collapsed means not uncollapsed.

Proof Intuition

The exact-distance requirement in almost uncollapsedness is much weaker than the large-distance requirement in uncollapsedness. A periodic map into the unit circle separates all pairs at one chosen distance, for instance distance π, but repeats values at arbitrarily large distances. This gives almost uncollapsedness and collapsedness at the same time. Since the circle is bounded, the same map is automatically not solvent. If one wants the domain to be infinite-dimensional, the same idea can be made set-theoretic: identify points modulo a large-period cyclic subgroup and send different cosets to different unit vectors.

Result

Theorem 1. *There are Banach spaces X, Y and a bounded map $f : X \to Y$ which is collapsed and almost uncollapsed. In particular, Problem 2.6 of [1] has an affirmative answer. Moreover, if the problem is read as asking for an infinite-dimensional domain, X can be chosen infinite-dimensional.*

Proof. Take $X = \mathbb{R}$, take $Y = \mathbb{R}^2$ with its Euclidean norm, and define

$$f(x) = (\cos x, \sin x).$$

The map is bounded, since $\|f(x)\|_2 = 1$ for every $x \in \mathbb{R}$.

It is almost uncollapsed. Indeed, if $|x - y| = \pi$, then $y = x + \pi$ or $y = x - \pi$, and hence $f(y) = -f(x)$. Therefore

$$\|f(x) - f(y)\|_2 = 2$$

for every pair $x, y \in \mathbb{R}$ with $|x - y| = \pi$. Thus

$$\overline{\rho}_f(\pi) = 2 > 0.$$

It is collapsed. Let $t > 0$. Choose an integer $k \geq 1$ such that $2\pi k \geq t$. Then $|2\pi k - 0| \geq t$, while

$$f(2\pi k) = f(0).$$

Consequently $\rho_f(t) = 0$ for every $t > 0$. Thus f is not uncollapsed, i.e. it is collapsed in the source terminology.

Finally, f is not solvent. Its image has diameter at most 2, so for any $n > 2$ no interval of domain distances can force all image distances to be greater than n. This proves the claimed bounded collapsed almost uncollapsed example.

For the infinite-dimensional variant, let X be any infinite-dimensional Banach space and choose $u \in X$ with $\|u\| = 1$. Let

$$A = 2\pi\mathbb{Z}u$$

and let $I = X/A$ be the set of cosets of the additive subgroup A. Put $Y = \ell_2(I)$, with its standard unit vectors $(e_i)_{i \in I}$, and define

$$F(x) = e_{x+A}.$$

This map is bounded, since every value has norm 1. If $\|x - y\| = \pi$ and $x + A = y + A$, then $x - y = 2\pi ku$ for some $k \in \mathbb{Z}$, so $\|x - y\| = 2\pi|k|$, impossible. Hence exact-distance-π pairs always belong to different cosets, and therefore

$$\|F(x) - F(y)\|_2 = \sqrt{2} \qquad (\|x - y\| = \pi).$$

Thus $\overline{\rho}_F(\pi) = \sqrt{2} > 0$. On the other hand, for every $t > 0$ one can choose $k \geq 1$ with $2\pi k \geq t$, and then

$$\|2\pi ku - 0\| \geq t, \qquad F(2\pi ku) = F(0).$$

So $\rho_F(t) = 0$ for every $t > 0$, and F is collapsed. Its image has diameter $\sqrt{2}$, so it is not solvent. This variant is not claimed to be continuous; it is included only to remove any possible finite-dimensional reading concern. □

Verification notes

The first proof uses only the source definitions and the elementary identities $f(x + \pi) = -f(x)$ and $f(x + 2\pi k) = f(x)$. The infinite-dimensional variant uses only cosets of the cyclic subgroup $2\pi\mathbb{Z}u$. The script `code/check_circle_map.py` performs a finite numerical smoke check of these identities and the boundedness normalization; it is not needed for the proof. The script `code/make_open_problem_crop.py` recreates the source screenshot from `source_paper.pdf`.

Verifier focus: confirm that the paper uses "collapsed" as the negation of "uncollapsed", as indicated in the surrounding text before Proposition 2.5, and that Problem 2.6 does not impose any additional regularity such as being a cocycle or a coarse map. The finite-dimensional displayed example is in fact Lipschitz, but it is not coarse-embedding-like because it is bounded. The infinite-dimensional variant answers only the unrestricted-map reading.

Novelty and literature check

Local duplicate check on June 14, 2026 searched `registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, and `proof_gaps/index.tsv` for `1607.06127`, title keywords, `collapsed`, `almost uncollapsed`, `bounded`, and related terms; no matching packet or attempt was found.

Web searches on June 14, 2026 used exact and close phrase searches for:

- collapsed, almost uncollapsed, bounded map, and Banach spaces;
- the source title together with almost uncollapsed and collapsed;
- the exact Problem 2.6 wording and shorter variants.

They found the source paper but no separate later paper explicitly answering this problem. Novelty confidence is moderate rather than high because the example is elementary and may be folklore.

Human review recommendation

Send to human as a likely valid full answer to Problem 2.6. The main semantic point to check is whether the author intended an unstated restriction to cocycles or to another class of maps; under the printed statement, the circle map answers the question.

References

[1] B. M. Braga, *On weaker notions of nonlinear embeddings between Banach spaces*, arXiv:1607.06127, 2016.